Lecture 9. Hypothesis Testing (Part 1)

One-Sample Tests and Weighted K-Sample Log-Rank Tests

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1 Overview

1.1 One-sample problems

Rationale. A special pattern in a KM curve may suggest a parametric model; parametric models are useful (efficiency; handling more complex sampling schemes), but once adopted we must check whether the model fits the data adequately. Hypotheses:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with S_0, H_0, h_0 *known*. E.g. under $X \sim \text{Exp}(\lambda)$: $H_0(t) = \lambda t$, $h_0(t) = \lambda$, $S_0(t) = e^{-\lambda t}$.

1.2 Two- and K-sample problems

The study population may be heterogeneous, with groups or key factors relevant to survival. With two groups we compare

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

equivalent hypotheses with *no* known form. The K-sample case is the obvious extension: $H_0 : S_1(t) = \cdots = S_K(t)$, $t \in [0, \tau]$.

1.3 How to test?

Without censoring: one-sample t test or Kolmogorov–Smirnov; two-/K-sample t tests and ANOVA (approximately normal data) or Wilcoxon rank-sum and Kruskal–Wallis (skewed data). The difficulty is censoring (and truncation); our focus is censoring.

2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$, X_i with hazard $h(t)$. Test $H_0 : h(t) = h_0(t)$, equivalently $H_0 : H(t) = H_0(t)$, for $t \in [0, \tau]$, where τ is typically the largest observed study time. Notation: $t_1 < \cdots < t_D$ the distinct death times, $d_j = \sum_i I(T_i = t_j, \delta_i = 1)$, $Y_j = Y(t_j) = \sum_i I(T_i \ge t_j)$.

Definition 9.1: One-sample weighted statistic

Let $W(t)$ be a weight function with $W(t) = 0$ whenever $Y(t) = 0$. Compare the Nelson–Aalen increments with the hypothesized hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under H_0, its variance is

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and for large samples $Z(\tau)^2 / V[Z(\tau)] \sim \chi^2_1$.

Remark 9.2: Intuition

$\tilde{H}(t) = \sum_{t_j \le t} d_j / Y_j$ estimates $H(t)$; the statistic is a weighted distance $\int W \, \mathrm{d}(\tilde{H} - H_0)$. Each increment d_j / Y_j has (Poisson) variance $\approx h_0 \, \mathrm{d}s / Y$, which gives $V[Z]$.

- The χ^2 statistic only serves two-sided alternatives.
- One-sided: for $H_A : h(t) > h_0(t)$ for some $t \le \tau$, reject for large positive $T_1 = Z(\tau) / \sqrt{V[Z(\tau)]}$; for $H_A : h(t) < h_0(t)$, reject for large negative T_1.

2.1 One-sample log-rank test

Theorem 9.3: $W(t) = Y(t)$: observed vs. expected

Allow left truncation: T_i = time on study, L_i = delayed entry time. With τ the largest time on study,

$$O(\tau) = \sum_{j=1}^{D} Y(t_j) \frac{d_j}{Y(t_j)} = \sum_{j=1}^{D} \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} \int_0^\tau I(L_i \le s \le T_i) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} [H_0(T_i) - H_0(L_i)].$$

O = observed number of events at or before τ; E = expected number under H_0. Also $V[Z(\tau)] = \int_0^\tau Y(s)h_0(s)\,\mathrm{d}s = E(\tau)$, so the test statistic is $(O-E)^2/E$.

2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t)S_0(t)^p[1-S_0(t)]^q, \qquad p,q \geq 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: more weight on **early** departures from H_0;
- $q \gg p$: more weight on **late** departures;
- $p = q$: more weight on departures in the **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

Example 9.4: IBCSG Trial 6

International Breast Cancer Study Group: a multi-center trial of adjuvant chemotherapy after surgical removal of the tumour; 735 subjects. The NA cumulative hazard plotted against time is roughly a straight line up to 100 months, suggesting an exponential distribution; eyeballing the slope gives $\hat{\lambda} \approx 0.005$. Formal one-sample log-rank tests ($W = Y$):

H_0	$T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$	p-value
$H_0(t) = 0.0045t$	-0.002	0.998
$H_0(t) = 0.004t$	1.8	0.07
$H_0(t) = 0.005t$	-1.4	0.16
$H_0(t) = 0.006t$	-4.1	< 0.01

$H_0(t) = 0.0045t$ fits best. (Sign check: a larger hypothesized rate inflates E, making $O - E$ negative.)

3 K-Sample Tests

3.1 Hypotheses

$$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \ \ \forall t \leq \tau \qquad \text{vs.} \qquad H_A : \text{ at least one } h_k(t) \text{ differs for some } t \leq \tau.$$

Equivalently in terms of H_k or S_k. Here τ is the largest time at which all groups have at least one subject at risk, typically $\tau = \min_k t_{\max,k}$. The alternative is *global*: reject if at least one population differs from the others at some time.

3.2 Data and notation

Right-censored data $\{(T_{ik}, \delta_{ik}),\ k = 1,\ldots,K,\ i = 1,\ldots,n_k\}$. Pool all data and let $0 < t_1 < \cdots < t_D$ be the distinct death times in the pooled sample:

- d_j, d_{jk}: deaths at t_j in the pooled sample / in group k;
- Y_j, Y_{jk}: number at risk at t_j in the pooled sample / in group k.

3.3 Basic idea and the test statistic

Under H_0 the hazards are equal but unknown, so both the pooled and within-group NA estimators estimate the same H:

$$\tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \le t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Build a test from a weighted difference between $\tilde{H}_k$ and $\tilde{H}$.

Definition 9.5: Weighted K-sample statistic

With $W_k(t) \ge 0$ a random weight, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with $W(\cdot)$ common to all groups, giving

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

3.4 Distribution under H_0

Theorem 9.6: Asymptotic normality (via martingale theory)

Under H_0, $(Z_1(\tau), \ldots, Z_K(\tau)) \approx \mathrm{MVN}(\mathbf{0}, \Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k = 1, \ldots, K,$$

$$\hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k \ne k'.$$

Derivation: hypergeometric origin of the variance

At t_j, condition on the margins: Y_j at risk split into $(Y_{j1}, \ldots, Y_{jK})$ with d_j deaths. Under H_0 the deaths are a random draw of d_j out of Y_j without replacement, so $(d_{j1}, \ldots, d_{jK})$ is multivariate hypergeometric with

$$\mathrm{E}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}, \quad \mathrm{Var}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}, \quad \mathrm{Cov}(d_{jk}, d_{jk'}) = -d_j\frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}.$$

The factor $(Y_j - d_j)/(Y_j - 1)$ is the finite-population correction (it equals 1 when $d_j = 1$). Summing over (approximately uncorrelated) death times gives $\hat{\sigma}_{kk'}$.

3.5 Linear dependency and the final statistic

Theorem 9.7: $\sum_k Z_k(\tau) = 0$

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j) \sum_{k=1}^{K} \Big\{ d_{jk} - d_j \frac{Y_{jk}}{Y_j} \Big\} = \sum_{j=1}^{D} W(t_j)(d_j - d_j) = 0,$$

since $\sum_k d_{jk} = d_j$ and $\sum_k Y_{jk} = Y_j$. Hence $Z_1, \ldots, Z_K$ are linearly dependent and Σ is singular.

Eliminate the redundancy by dropping one (arbitrary) $Z_K(\tau)$:

$$(Z_1, \ldots, Z_{K-1}) \approx \text{MVN}(\mathbf{0}, \Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau), \ldots, Z_{K-1}(\tau))\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, (Z_1(\tau), \ldots, Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject at level α if $\chi^2 > \chi^2(K-1, 1-\alpha)$. (The value does not depend on which Z_k is dropped.)

3.6 Special case $K = 2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat{V}(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1} - Y_{j1} d_j / Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j) \frac{Y_{j1}}{Y_j}(1 - \frac{Y_{j1}}{Y_j}) \frac{Y_j - d_j}{Y_j - 1} d_j}} \overset{H_0}{\approx} N(0,1).$$

For $K = 2$, a two-sided level-α test rejects if $|Z| > Z_{1-\alpha/2}$; one-sided tests are possible. For $K > 2$ we must use the χ^2 test, which only handles two-sided alternatives.

4 Choice of the Weight Function

Different weights change the power against specific alternatives; they are extremely well studied. All have the form $W_k(t_j) = Y_{jk} W(t_j)$.

1. **Log-rank**: $W(t) = 1$. Optimal power against **proportional hazards** alternatives $h_1(t) = c_2 h_2(t) = \cdots = c_K h_K(t)$.
2. **Gehan** (generalized Wilcoxon): $W(t) = Y(t) = \sum_k \sum_i I(T_{ik} \geq t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$. Commonly used.
3. **Tarone–Ware**: $W(t) = f(Y(t))$ with known increasing f; $f(u) = u$ is Gehan, the original Tarone–Ware uses $f(u) = u^{1/2}$, also $f(u) = \log u$. Gives relatively larger weight to death times with larger risk sets.
4. **Peto–Peto**: $W(t) = \tilde{S}(t)$, $\tilde{S}(t) = \prod_{t_j \leq t} (1 - \frac{d_j}{Y_j + 1})$, a KM-like estimator from the pooled sample.
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t) Y(t)/(Y(t)+1)$. Both Peto–Peto tests depend on the pooled survival function.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p [1 - \hat{S}(t_{j-1})]^q$, $p, q \geq 0$, $\hat{S}$ the pooled KM.
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: larger weights early $\Rightarrow$ sensitive to early differences among hazards;
   - $p = 0, q > 0$: larger weights late $\Rightarrow$ sensitive to late differences.

Remark 9.8: Why the left-continuous $\hat{S}(t_{j-1})$

The weight at t_j must be *predictable* — known just before t_j — so it uses survival information up to t_{j-1}, not the deaths at t_j themselves. This is what makes Z_k a martingale-based statistic.

5 Catheter Placement in Kidney Dialysis Patients

Clinical trial (K&M Section 1.4) comparing time to cutaneous exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2). From the KM curves (K&M Fig. 7.1), the surgical group does better early, but the percutaneous group has higher infection-free probability later (about 0.8 vs. 0.2 near month 27). The curves *cross*, so the choice of weight matters:

- weighting early times heavily (e.g. Gehan) may suggest the surgical group is better;
- weighting late times heavily may suggest the percutaneous group is better.

Relative weights (K&M Fig. 7.2): Gehan and Tarone–Ware decrease steeply with time; log-rank is flat; Fleming–Harrington with $q > 0$ increase with time.

Example 9.9: Log-rank computation (K&M Table 7.2, selected rows)

t_j	Y_{j1}	d_{j1}	Y_{j2}	d_{j2}	Y_j	d_j	$Y_{j1}\frac{d_j}{Y_j}$	$d_{j1} - Y_{j1}\frac{d_j}{Y_j}$	$\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$
0.5	43	0	76	6	119	6	2.168	−2.168	1.326
1.5	43	1	60	0	103	1	0.417	0.583	0.243
2.5	42	0	56	2	98	2	0.857	−0.857	0.485
4.5	36	2	43	0	79	2	0.911	1.089	0.490
							⋮		
26.5	2	1	3	0	5	1	0.400	0.600	0.240
SUM		15		11		26	11.036	3.964	6.211

So $Z_1(\tau) = 15 - 11.036 = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 3.964^2/6.211 = 2.53$, $p = 0.112$: no significant difference by the log-rank test.

Example 9.10: Comparison of two-sample tests (K&M Table 7.3)

Test	$W(t_j)$	$Z_1(\tau)$	σ^2_{11}	χ^2	p-value
Log-rank	1	3.96	6.21	2.53	0.112
Gehan	Y_j	−9	38862	0.002	0.964
Tarone–Ware	$Y_j^{1/2}$	13.20	432.83	0.40	0.526
Peto–Peto	$\tilde{S}(t_j)$	2.47	4.36	1.40	0.237
Modified Peto–Peto	$\tilde{S}(t_j)Y_j/(Y_j+1)$	2.31	4.20	1.28	0.259
FH $p=0, q=1$	$1-\hat{S}(t_{j-1})$	1.41	0.21	9.67	0.002
FH $p=1, q=0$	$\hat{S}(t_{j-1})$	2.55	4.69	1.39	0.239
FH $p=1, q=1$	$\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$	1.02	0.11	9.83	0.002
FH $p=0.5, q=0.5$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$	2.47	0.66	9.28	0.002
FH $p=0.5, q=2$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^2$	0.32	0.01	8.18	0.004

Remark 9.11: Comments

- Gehan's weight puts very heavy weight on early comparisons, leading to a *negative* $Z_1(\tau)$.
- Fleming–Harrington tests with $q > 0$ weight late comparisons and are significant, because the two curves diverge for large t.
- In practice, the choice of weight is an important decision and should be made *before* looking at the data.
- The log-rank test ($W = 1$) is the most popular, followed by Gehan (generalized Wilcoxon rank-sum); both are in most statistical packages.

6 Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival of leukaemia patients after bone marrow transplantation: Group 1 = 38 ALL, Group 2 = 54 AML low-risk, Group 3 = 45 AML high-risk. Test $H_0 : S_1(t) = S_2(t) = S_3(t)$, $t \le \tau = 2204$ days.

Example 9.12: Log-rank test with $K = 3$

Using the first two statistics, $Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$, with

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1} \begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On $K - 1 = 2$ df, $p \approx 0.001$.

Derivation: evaluating the quadratic form

$\det \hat{\Sigma} = 15.9552(20.3398) - 10.3451^2 = 324.53 - 107.02 = 217.51$, and $\hat{\Sigma}^{-1} = \frac{1}{217.51}\begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\begin{aligned} \chi^2 &= \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big] \\ &= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80. \end{aligned}$$

Other weights: Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 ($p = 0.0004$); FH $(1, 0)$ 15.67 ($p = 0.0004$); FH $(0, 1)$ 6.11 ($p = 0.047$); FH $(1, 1)$ 9.93 ($p = 0.007$). The results are robust: all tests suggest that the disease-free survival curves differ among the three groups.